Hermite–Padé approximations for Nikishin systems and the irrationality of $\zeta(3)$

V.N. Sorokin

1. In 1978 Apéry [1] proved the irrationality of the number $\zeta(3)$, where $\zeta(s)$ is the Riemann zeta function. Namely, he produced rational integers q_n and p_n, where $n \in \mathbb{Z}_+$, such that $r_n = q_n 2\zeta(3) - p_n \neq 0$ and $r_n \to 0$ as $n \to \infty$. In [2] Beukers showed that these numbers are values of some Hermite–Padé approximations for polylogarithms. A similar construction shows the irrationality of $\zeta(2)$ as well. Although the last assertion is well-known: $\zeta(2) = \pi^2/6$, nevertheless, the construction allows us to obtain a good quantitative estimate of the measure of irrationality of this number. We have studied the following question: what other Hermite–Padé approximations lead to a proof of irrationality of $\zeta(2)$ and $\zeta(3)$? As a result, we have obtained, besides the Beukers constructions, still another construction for $\zeta(2)$ and three for $\zeta(3)$. In this paper we concentrate on one of these six problems.

2. We consider the power series

$$e(x) = \sum_{n=1}^{\infty} \frac{1}{n} \frac{1}{x^n}, \quad d(x) = \sum_{n=1}^{\infty} \frac{1}{n^2} \frac{1}{x^n}, \quad h(x) = \sum_{n=1}^{\infty} \frac{1}{n} \sum_{k=n}^{\infty} \frac{1}{k^2} \frac{1}{x^n},$$

and also

$$\varepsilon(x) = \sum_{n=1}^{\infty} \frac{1}{n}(1-x)^n, \quad \varphi(x) = \sum_{n=1}^{\infty} \frac{1}{n} \sum_{m=1}^{n} \frac{1}{m}(1-x)^n.$$

(*h*) *Let* $n \in \mathbb{Z}_+$. *Find polynomials* $A_n \neq 0$, B_n, C_n, D_n *of degrees not exceeding* $n, 2n-1, 2n, 2n-1$, *respectively, for which*

$$h_n(x) = A_n(x)h(x) + \{B_n(x)(\zeta(2) - d(x)) + C_n(x)e(x) + D_n(x)\}/(1-x)^n = O(1/x^{n+1}),$$
$$\varphi_n(x) = A_n(x)\varphi(x) + \{-B_n(x)\varepsilon(x) + C_n(x)\}/(1-x)^n = O((1-x)^{n+1}),$$
$$e_n(x) = A_n(x)e(x) + B_n(x)/(1-x)^n = O(1/x^{n+1}),$$
$$N_n(x) = -B_n(x)\varphi(x) + D_n(x) = O((1-x)^n).$$

Problem (*h*) has a unique (up to normalization) solution. We choose the normalization such that $A_n(0) = (-1)^n$. We denote by ω_n the l.c.m. of the numbers $1, 2, \ldots, n$. Then the Apéry approximation of $\zeta(3)$ is $r_n = \omega_n^3 h_n(1)$. Then $r_n^{1/n} \to e^3(\sqrt{2}-1)^4 = 0.59\ldots$.

3. The functions h, d, e form an MT-system of Nikishin [3]. It follows from the orthogonality relations for MT-systems that each of the functions A_n, e_n, φ_n has n simple zeros in the interval $(0,1), (-\infty, 0), (0,1)$, respectively. The limit distributions of these zeros exist, they are probability measures $\mathfrak{a}, \mathfrak{e}, \mathfrak{f}$.

We shall use the notation: $S(\mathfrak{f})$ is the support of the measure $\mathfrak{f}$, $V_{\mathfrak{f}}$ is the logarithmic potential of the measure $\mathfrak{f}$, and δ is the unit Dirac measure at the point $x = 1$.

The following result can be obtained by a method of potential theory [4].

***Theorem* 1.** *The measures* $\mathfrak{a}, \mathfrak{e}, \mathfrak{f}$ *form the unique solution of the equilibrium problem*

$$(2V_{\mathfrak{a}} - V_{\mathfrak{e}} + V_{\delta})(x) \quad \begin{cases} = w_{\mathfrak{a}}, & x \in S(\mathfrak{a}), \\ \geq w_{\mathfrak{a}}, & x \in [0,1] \setminus S(\mathfrak{a}), \end{cases}$$

$$(2V_{\mathfrak{e}} - V_{\mathfrak{a}} - V_{\mathfrak{f}} - 2V_{\delta})(x) \quad \begin{cases} = w_{\mathfrak{e}}, & x \in S(\mathfrak{e}), \\ \geq w_{\mathfrak{e}}, & x \in (-\infty, 0] \setminus S(\mathfrak{e}), \end{cases}$$

$$(2V_{\mathfrak{f}} - V_{\mathfrak{e}} + V_{\delta})(x) \quad \begin{cases} = w_{\mathfrak{f}}, & x \in S(\mathfrak{f}), \\ \geq w_{\mathfrak{f}}, & x \in [0,1] \setminus S(\mathfrak{f}), \end{cases}$$

where $w_{\mathfrak{a}}, w_{\mathfrak{e}}, w_{\mathfrak{f}}$ *are constants.*

4. For the solutions of problem (h) we have obtained explicit formulae. We mention only two of them.

***Theorem* 2.** 1) *We have the Rodrigues formula*

$$A_n(x)=\frac{1}{n!}\left(\frac{d}{dx}\right)^n x^n(1-x)^{2n+1}\frac{1}{n!}\left(\frac{d}{dx}\right)^n\frac{x^n}{(1-x)^{n+1}}\frac{1}{n!}\left(\frac{d}{dx}\right)^n x^n(x-1)^n.$$

2) *We have the integral representation*

$$h_n(x)=\int_0^1\int_0^1\int_0^1\frac{u^n(1-u)^n v^n(1-v)^n w^n(1-w)^n\,dudvdw}{(x-u(x-v(x-w)))^{n+1}}.$$

5. By applying the saddle-point method to the formulae in section 4 we study the asymptotic behaviour of solutions of problem (h) as $n\to\infty$. We restrict ourselves to formulae of weak asymptotics.

We consider the equation

$$(*)\quad (1-x)u^4+(44x^2-52x+4)u^3+(16x^3-40x^2+154x+6)u^2+(44x^2-52x+4)u+(1-x)=0.$$

The Riemann surface of the algebraic function $u(x)$—the solution of equation $(*)$—is constructed in the following manner. In the region $\mathbb{C}\setminus[-1,4/5]$ we choose two univalent branches: u_e and $u_\varphi=1/u_e$ such that $u_e(x)\sim\theta^5/(4x)$ as $x\to\infty$, where $\theta=(1+\sqrt{5})/2$ is the golden ratio. In the region $\mathbb{C}\setminus[0,4/5]$ we choose two univalent branches: u_a and $u_h=1/u_a$ such that $u_a(x)\sim 4\theta^5 x$ as $x\to\infty$. On the cut $[0,4/5]$ we "glue" the branches u_e and u_a, and also u_φ and u_h, and on the cut $[-1,0]$ we "glue" the branches u_φ and u_e.

The equation $(*)$ can be solved easily with radicals, namely:

$$u_\varphi=1/u_e=p_+-\sqrt{p_+^2-1},\qquad u_a=1/u_h=p_--\sqrt{p_-^2-1},$$

where

$$p_\pm(x)=\left(-(11x^2-13x+1)\pm\sqrt{x(5x-4)^3}\right)\Big/(1-x),$$

and the branches of the radicals are chosen in this way: $\sqrt{x(5x-4)^3}\sim 5\sqrt{5}x^2$ and $\sqrt{p_\pm^2(x)-1}\sim p_\pm(x)$ as $x\to\infty$.

***Theorem* 3.** *The following asymptotic formulae hold uniformly inside the indicated regions*:

$$\lim_{n\to\infty}|A_n(x)|^{1/n}=|u_a(x)|,\qquad x\in\mathbb{C}\setminus[0,4/5],$$

$$\lim_{n\to\infty}|e_n(x)|^{1/n}=|u_e(x)|,\qquad x\in\mathbb{C}\setminus[-1,4/5],$$

$$\lim_{n\to\infty}|\varphi_n(x)|^{1/n}=|u_\varphi(x)|,\qquad x\in\mathbb{C}\setminus[-1,4/5],$$

$$\lim_{n\to\infty}|h_n(x)|^{1/n}=|u_h(x)|,\qquad x\in\mathbb{C}\setminus[0,4/5].$$

As a corollary we deduce that the measures α and $\mathfrak{f}$ coincide, their support is the interval $[0,4/5]$, $S(\mathfrak{e})=[-1,0]$, and the equilibrium constants are $w_\alpha=w_\mathfrak{f}=4\ln 2$ and $w_\mathfrak{e}=10\ln\theta-4\ln 2$. The equilibrium measures can be expressed easily in terms of the roots of equation $(*)$.

References

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Moscow Institute of Radiotechnology, Electronics, and Automation

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